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GST 104: History and Philosophy of Science - How Knowledge Actually Grows

From observation and falsification to African science and the ethics of discovery

GST / COMPULSORY

GST & Compulsory Courses

General Studies (GST / CED)
GST Directorate | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

Study smarter. Ask better questions. Pass with understanding.

Independent study support | Not an official UNIZIK publication

START HERE

Why this pack exists

This pack explains what makes science different from mere belief, how scientific ideas have changed across history, and how African knowledge systems contributed to medicine, metallurgy and agriculture. It also gives you the vocabulary for ethics-of-science questions.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Explain the scientific method, hypothesis, falsifiability, paradigm and theory without collapsing them into one word.
- Trace major shifts from ancient to modern science with accurate examples.
- Describe African contributions to science and technology using specific evidence.
- Discuss ethical issues such as research misconduct, informed consent and environmental responsibility.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

What science is - and is not

Science is a method of producing reliable knowledge through observation, measurement, hypothesis testing and public criticism. It is not a fixed body of facts, and it is not the same as technology.

A hypothesis is a testable prediction. A theory is a well-supported explanatory framework that has survived repeated testing. A law describes a regular relationship. Colloquial English uses 'theory' to mean a guess; science does not.

Falsifiability, associated with Karl Popper, means a claim must specify what evidence would count against it. 'Everything happens for a reason' is not scientific because no observation could refute it.

- Observation and measurement: record what happens under controlled conditions.
- Hypothesis: a specific, testable prediction.
- Experiment: manipulate variables while controlling others.

- Peer review: allow others to check methods, data and reasoning.
- Replication: repeat the study to see whether the result holds.

> **Correlation is not causation**
> Two things rising together may share a third cause, or one may cause the other in the opposite direction. Before claiming causation, look for controlled evidence and a plausible mechanism.

A short history of scientific change

Science has not progressed by simply accumulating facts. It has changed when old frameworks could no longer explain new observations.

- Ancient Greek natural philosophy: Aristotle's physics and the geocentric model.
- Islamic Golden Age: algebra, optics, hospitals and the preservation of Greek texts.
- Copernican revolution: the sun-centred model replaced geocentrism.
- Galileo and Newton: mathematised motion, telescopes and laws of gravitation.
- Darwin and Wallace: evolution by natural selection.
- Einstein: relativity transformed space, time and gravity.
- Watson, Crick, Franklin and Wilkins: the structure of DNA and modern molecular biology.

African science and technology

African societies produced empirical knowledge and technical innovation long before European contact. Recognising this is not sentiment; it is historical accuracy.

- Metallurgy: Nok iron smelting and Igbo-Ukwu bronze casting.
- Medicine: herbal pharmacopoeias, bone setting and knowledge of plant properties, some of which inform modern drug discovery.
- Agriculture: mixed cropping, soil management, terracing and indigenous seed selection.
- Mathematics: fractal-like patterns in African textiles and architecture; counting and weight systems.
- Astronomy: navigation and calendar systems based on celestial observation.

> **Avoid both extremes**
> Do not claim that all modern science was stolen from Africa, and do not claim that Africa had no science. Make specific, evidence-based claims about specific technologies.

Science, technology and society

Scientific knowledge is never automatically good. Its application depends on incentives, regulation and who bears the risk.

- Research ethics: informed consent, confidentiality, honest reporting and approval by an ethics committee.
- Misconduct: fabrication, falsification and plagiarism destroy trust and can harm patients.
- Dual use: the same research can produce medicine or weapons.

- Environmental justice: communities may bear environmental costs without sharing the benefits.
- Access: a drug that exists but is unaffordable is not a solution for the person who needs it.

PRACTICE, NOT PREDICTION

Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. What makes a claim falsifiable?

Answer: It specifies observations that would count against it. If no possible evidence could refute the claim, it is not scientifically testable.

2. What is the difference between a hypothesis and a theory?

Answer: A hypothesis is a specific testable prediction. A theory is a broad explanatory framework supported by many lines of evidence.

3. Give one African contribution to metallurgy.

Answer: The Nok culture produced iron smelting evidence dating to roughly the first millennium BCE; Igbo-Ukwu is known for sophisticated bronze casting.

4. Why is peer review important?

Answer: It exposes methods and claims to independent scrutiny before publication, reducing the chance that errors or weak reasoning go unchallenged.

5. Explain the difference between correlation and causation with an example.

Answer: Ice-cream sales and drowning deaths both rise in hot weather, but ice cream does not cause drowning. Heat is a common cause.

6. Name two ethical requirements for research involving human participants.

Answer: Informed consent and ethics committee approval; confidentiality and the right to withdraw are also essential.

Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] Write definitions of hypothesis, theory, law and paradigm in your own words.
- [ ] Create a timeline of six scientific shifts with one key figure each.
- [ ] List four African technologies and the evidence that supports them.
- [ ] Find one news report that confuses correlation with causation and rewrite the headline accurately.
- [ ] Write a short paragraph on why falsifiability matters.
- [ ] Explain informed consent to a friend in under one minute.
- [ ] Review one unethical research case and identify what should have been done differently.

Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

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YOUR WORKSPACE

Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

Three ideas I can now explain without notes:

A question I still cannot answer:

For my lecturer or course rep:

Next revision action and date:

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